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Feature Representation, Cluster Structure and Explanation Tradeoffs in Customer Segmentation

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Abstract

Customer segmentation depends on the information used to describe each customer, yet representation is often treated as neutral. This dissertation compares a compact transaction representation with a nested twelve-dimensional behavioural representation for the same 11,719 RetailRocket purchasers. The cohort and evaluation design are held constant. CLASSIX supplies intrinsic aggregation provenance, while ExKMC supplies post hoc rules for K Means assignments.

Changing the representation produces markedly different partitions. Independent full-data selection gives ARI 0.014 for the two four-cluster K Means results and ARI 0.068 for CLASSIX. Across 41 CLASSIX settings acceptable in both spaces, matched-parameter ARI ranges from -0.051 to 0.392. Mean fixed-parameter overlap-resampling ARI remains above 0.964, showing that a configuration can be repeatable within each representation while the representations disagree.

The explanation tradeoff also changes. At a common four-leaf budget, compact ExKMC trees reach mean held-out fidelity 0.9448 and rich trees reach 0.9650 with the same depth. At sixteen leaves the values are 0.9924 and 0.9747. Rich CLASSIX produces 559 aggregation groups compared with 131 in the compact fit. One of four rich K Means clusters passes the documented addressability audit. No compact K Means cluster or CLASSIX cluster passes.

Representation changes observable cluster structure and the conditions needed to explain it. Richer behaviour does not automatically improve internal quality or produce deployment-ready segments. Its business value remains limited to descriptive and queryable information until prospective intervention outcomes are measured.

Declaration

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AI Statement

Artificial intelligence tools were used as assistants for code drafting, data processing, numerical checks, source discovery and English language polishing. The author determined the research design, verified all sources and results, reviewed every substantive claim and accepts responsibility for the report. Details relevant to reproducibility are recorded in the accompanying materials.

Chapter 1

Introduction

1.1 Research background

Customer segmentation converts behavioural records into groups that can be inspected, compared and queried. A clustering algorithm never observes a customer directly. It receives a vector assembled from selected events, time windows and aggregation rules. A compact purchase summary emphasises transaction timing and repetition. A rich event representation also records browsing volume, conversion ratios, temporal engagement and category breadth. These choices define different distances between the same people and can make different cluster structures visible.

Representation is often evaluated indirectly through the score of the final clustering. That approach cannot separate information choice from algorithm behaviour when cohorts, parameters or validation rules also change. It can also favour a high silhouette partition that isolates a few unusual observations while leaving almost every customer in one group. Clusterability research shows that structure depends on the adopted definition and data conditions rather than on a universal test (Ackerman and Ben-David, 2009; Adolfsson, Ackerman and Brownstein, 2019). Representation must be examined as part of the empirical model for customer event logs.

Explanation adds a second source of ambiguity. CLASSIX provides intrinsic provenance through sorted aggregation groups, starting points and merges (Chen and Güttel, 2024). A threshold tree for K Means instead provides a post hoc approximation whose rule count and fidelity can be measured (Dasgupta et al., 2020). These explanations refer to different clustering objects. A short tree may closely reproduce a reference partition without showing how that partition was discovered. A detailed intrinsic trace may reproduce every assignment while remaining difficult to convert into a small set of business conditions. Conflating the two would hide the effect that representation has on both structure and explanation burden.

1.2 Research purpose and significance

The purpose of this dissertation is to examine how a compact transaction representation and a rich behavioural representation affect clustering for the same 11,719 RetailRocket purchasers. The compact representation contains purchase recency, purchase occasions and purchase event count. The rich representation retains those variables and adds nine activity, conversion, temporal and category measures. Both representations use the same event snapshot, customer order, transformations, algorithm grids, deployment constraints and resampling seeds. The study therefore compares added behavioural information within one recorded population rather than comparing unrelated cohorts.

The significance of the study is methodological and bounded by this setting. Holding the cohort and evaluation rules fixed helps separate the effect of representation from changes in population or model selection. The analysis also distinguishes native CLASSIX provenance from post hoc ExKMC approximation because they explain different clustering objects. The mathematical discussion restates projection pruning for the installed CLASSIX algorithm and formulates a local sufficient stability condition.

1.3 Research content

The research begins with nine labelled datasets that check CLASSIX, K Means, DBSCAN and Ward under label-free selection. The main analysis then applies K Means and CLASSIX to both RetailRocket representations and compares cluster counts, internal indices, customer assignments and fixed-parameter overlap-resampling stability. CLASSIX provenance and held-out ExKMC trees provide separate explanation evidence. Feature deletion, PCA, perturbation, alternative session windows and permuted dimensions examine sensitivity. A mathematical analysis states the scope of projection pruning, distance complexity, determinism and local stability.

The selected K Means and CLASSIX partitions have cross-representation ARI values of 0.014 and 0.068. Compact spaces give higher silhouette values, while rich K Means gives more balanced clusters and rich CLASSIX gives more aggregation groups. At four leaves, ExKMC held-out fidelity is 0.9448 for compact K Means and 0.9650 for rich K Means. Only one rich K Means cluster passes the addressability audit. These findings show that added behaviour changes the observed grouping and explanation burden, but they do not establish that one representation is universally preferable or that the clusters have realised business value.

Chapter 2

Literature Review

Research on customer segmentation often treats the representation as a neutral input to clustering. That assumption is difficult to defend for event logs. A customer can be described by a few purchase summaries, the full sequence of recorded actions, or a rich set of temporal, conversion and category signals. Each choice changes the distances observed by the algorithm. Clickstream research describes raw action sequences as large, variable and noisy, then frames useful segments as subsets recovered through constraints on derived attributes (Dextras-Romagnino and Munzner, 2019). Work on object centred event logs likewise converts traces into vector profiles before clustering and pattern analysis (Faria Junior et al., 2023). Customer behaviour studies construct features for segmentation and later decision tasks, but their application specific choices do not establish a universal representation (Abdi and Abolmakarem, 2019). Representation is part of the empirical model rather than a preliminary clerical step.

Clustering does not reveal a representation independent truth. Clusterability theory supplies several incompatible formal notions of when data possess recoverable structure (Ackerman and Ben-David, 2009). Empirical comparisons also show that clusterability measures depend on distributional conditions, dimension and the type of structure sought (Adolfsson, Ackerman and Brownstein, 2019). A high internal score supports a partition within a defined feature space, not an intrinsic segmentation outside that space. Dimensional reduction retains this dependence. It replaces named variables with a new geometry and can obscure the link between a cluster and the original attributes (Bertsimas, Orfanoudaki and Wiberg, 2021). A paired comparison on the same customers is more informative than results drawn from unrelated populations.

Explainable clustering contains three distinct design positions. Intrinsic methods expose the computations that create a partition. CLASSIX sorts observations along a principal direction, aggregates nearby points and merges the resulting groups (Chen and Güttel, 2024). Its public group assignments, starting points and final labels support a trace from an observation to a cluster. Explainability first methods instead make a simple description part of the clustering objective. ICOT constructs tree partitions through optimisation, while XClusters balances distortion against the size and accuracy of a tree during cluster formation (Bertsimas, Orfanoudaki and Wiberg, 2021; Hwang and Whang, 2023). Post hoc methods begin with a reference partition and approximate it with rules or a tree. These positions explain different objects. Provenance from an intrinsic algorithm answers how its own grouping arose. A post hoc tree answers how closely axis aligned rules can reproduce another partition. Neither answer can be converted into the other without changing the target of explanation.

Table 2.1 Comparison of explanation designs in the reviewed literature

Method	Target	Timing	Form	Complexity control	Status
CLASSIX	Own partition	Intrinsic	Groups and starting points	Radius and minimum size	Journal
IMM	K Means objective	Tree construction	Threshold tree	One leaf per cluster	Conference
ExKMC	K Means labels	Post hoc	Expanded tree	Leaf budget	Preprint
CREAM	Fitted partition	Post hoc	Cubes and tree rules	Tuning settings	Conference
ICOT	Own partition	Joint	Optimised tree	Validation objective and tree	Journal
XClusters	Joint objective	Joint	Decision tree	Distortion and node tradeoff	Conference

Decision trees make the tradeoff between concise expression and partition quality explicit. IMM uses threshold cuts to construct a small tree and analyses the additional K Means or K Medians cost caused by that restriction (Dasgupta et al., 2020). Theoretical work shows that such simplicity can impose unavoidable objective loss and that the magnitude depends on the clustering problem, cluster count and dimension (Laber and Murtinho, 2021). Other work asks how a fixed clustering can be made explainable after removing a limited number of outlying observations, which further demonstrates that an apparently simple tree may obtain its clarity by changing the population being explained (Bandyapadhyay et al., 2022). CREAM constructs hypercubic approximations through decision tree induction and tunes the resulting approximation, extending the same concern beyond one reference algorithm (Sabbatini and Calegari, 2023). These methods make complexity measurable through leaves, depth, conditions and features, but those counts do not by themselves establish that a person will understand or use the explanation.

Theory and software maturity also vary across these approaches. IMM and price analyses provide approximation bounds under defined objective functions, whereas CLASSIX supplies a maintained implementation and public explanation attributes for its own partition. CREAM, ICOT and XClusters broaden the expression and optimisation choices, but their guarantees and computational demands concern different targets. Queryable rules, bounded objective loss and native provenance are complementary properties rather than grades on one explanation scale.

ExKMC expands a tree beyond one leaf for each K Means cluster, allowing the leaf budget to test whether additional rules improve agreement with a fixed reference. Its verified source is an arXiv preprint rather than a peer reviewed publication. It documents the implemented method but does not provide independent evidence of general effectiveness (Frost, Moshkovitz and Rashtchian, 2020). Fidelity here means exact predictive agreement between the tree and held out K Means assignments. It is neither an internal clustering metric nor evidence that the K Means partition is substantively correct. More general XAI research finds that fidelity measures can disagree even when a transparent model supplies a known reference, so the target and calculation must be stated directly (Miró-Nicolau, Jaume-i-Capó and Moyà-Alcover, 2025).

Operational usefulness introduces another boundary. A tree or a set of threshold conditions may be machine queryable, while an intrinsic trace may identify prototypes and merge relations. Queryability can be observed through the number of conditions, the use of named features, customer coverage and agreement with a reference partition. It does not establish improved revenue, better interventions or human comprehension. Segmentifier illustrates why recorded constraints and provenance can support the refinement and export of behavioural segments, but its design study does not validate the commercial effect of a clustering model (Dextras-Romagnino and Munzner, 2019). Similarly, theoretical guarantees on objective cost do not measure whether a segment is large enough, distinct enough or sufficiently covered by reliable attributes to be acted upon.

In line with these observations, the present study compares compact transaction and rich behavioural representations for the same RetailRocket purchasers under matched clustering and evaluation conditions, and relates differences in cluster structure to CLASSIX provenance, ExKMC fidelity and auditable addressability.

Chapter 3

Data and Methodology

This chapter explains how the RetailRocket event data are converted into two customer representations and evaluated under matched analytical conditions. It also distinguishes the mathematical properties of CLASSIX from the empirical robustness checks conducted in this study.

3.1 Research design

The research design compares two descriptions of the same purchasers. The compact representation contains three variables that summarise purchase timing and frequency. The rich representation contains twelve variables. It retains the same three purchase variables and adds nine measures of activity, conversion, temporal engagement and category behaviour. Both representations are constructed from the same event history for the same 11,719 customers. Their comparison therefore examines how the additional behavioural variables alter the geometry presented to the clustering algorithms.

3.1.1 Data source

The empirical comparison uses the RetailRocket recommender system dataset, an event log released under the Creative Commons Attribution NonCommercial ShareAlike licence (Retailrocket, no date). The data used in the analysis contain 2,756,101 event rows, two item property files with 20,275,902 rows in total, and a category tree with 1,669 nodes. Events record an anonymous visitor, time, item and action type. Transaction identifiers are present on purchase events, but price, expenditure and revenue are absent. Event frequency is not interpreted as monetary value, and purchase rows are not described as product quantities.

The study population contains every visitor with at least one transaction event. Applying this rule to the event table yields 11,719 purchasers and 230,678 event rows for those purchasers. Both representations use these same customers in the same order. The observation snapshot is one day after the final recorded event, which gives 19 September 2015 at 02.59.47 UTC. Purchase recency is measured from this snapshot rather than from each customer's final browsing action. Stable ordering by visitor, timestamp and original row position resolves equal timestamps without making an arbitrary event sequence depend on file reading behaviour.

3.1.2 Feature engineering

Three purchase summaries form the compact representation. Purchase recency is the elapsed number of days from the last transaction event to the snapshot. Purchase occasion count is the number of distinct nonempty transaction identifiers. Purchase event count is the number of transaction rows. The latter two measures can differ when several transaction rows share an identifier. They describe purchase activity recorded in the event log and do not substitute for expenditure.

Nine additional variables expand the compact representation into the rich representation. They describe activity volume, conversion, timing and category behaviour. Table 3.1 defines all twelve variables. The three purchase variables are carried into the rich table without recalculation, so each purchaser has the same purchase summary in both representations.

Table 3.1 Recorded feature contract for the two customer representations

Feature	Block	Operational definition
Purchase recency days	Compact transaction	Days from the latest transaction event to the snapshot
Purchase occasion count	Compact transaction	Distinct nonempty transaction identifiers
Purchase event count	Compact transaction	Transaction event rows
Total events	Activity	All event rows for the visitor
Active days	Activity	UTC calendar days containing an event
Cart per view	Conversion	Add to cart events divided by view events
Transaction per view	Conversion	Transaction events divided by view events

Feature	Block	Operational definition
Activity span days	Temporal engagement	Days between first and last event
Mean interevent hours	Temporal engagement	Mean gap between adjacent events
Session count 30m	Temporal engagement	Sessions separated by more than thirty minutes
Distinct top categories	Category behaviour	Distinct root categories among known joins
Top category share	Category behaviour	Largest root category count divided by known category events

Category variables require a time-valid join because item properties change during the observation period. Each event receives the latest category property that was available at or before its timestamp. The category tree then maps that category to its root. This procedure covers 2,099,173 event rows, which is 76.16 per cent of all events. Unknown joins are excluded from category denominators rather than treated as a category. A total of 1,168 purchasers have no event with a known category. Their two category features are set to zero, while a separate coverage indicator records the missing information. This indicator is used for data checks and is not included among the twelve clustering variables. The missing category records prevent these variables from being interpreted as complete purchase preferences.

Ratio variables also use explicit denominator rules. Cart per view and transaction per view equal zero when a visitor has no recorded view event. This condition affects 428 purchasers and is retained in the feature audit. A session starts with a visitor's first event or after an inactivity gap greater than thirty minutes. Alternative gaps of fifteen and sixty minutes are reserved for sensitivity analysis. These choices make every derived value reproducible while exposing the assumptions most likely to affect temporal structure.

Nonnegative counts and time variables receive a logarithm after adding one. Ratios and shares remain on their original scale. Each representation is then standardised feature by feature using its own fitted means and standard deviations. Full-data clustering fits these transformations on the complete cohort. Sample splitting and resampling fit them only on the relevant training or subsample rows. PCA is applied only to a separately standardised copy of the rich representation and never supplies features to rule explanations or cluster descriptions.

Before clustering, both feature tables were checked against the source data. File hashes identify the exact raw inputs. Unit tests recalculate features for small manually checked event histories that include equal timestamps, category changes, missing categories and zero denominators. Each table must contain the same 11,719 unique purchaser identifiers in the same order. The three purchase variables are then compared row by row and must be identical across the two tables. These checks ensure that the compact and rich analyses differ through the nine additional behavioural variables rather than through different customers or conflicting purchase summaries.

3.2 Analytical methods and mathematical scope

The comparison uses one recorded customer cohort, feature contract, set of candidate grids, deployment constraints and evaluation measures. K Means provides a centre-based reference partition that can be approximated by ExKMC rules. CLASSIX provides a separate clustering result with native aggregation provenance. ExKMC is not a clustering algorithm, and no composite score ranks it against CLASSIX. The same change in represented information is traced through each partition and its corresponding explanation.

3.2.1 Clustering candidates and selection

K Means uses k-means++ seeding, twenty initialisations and the fixed seed 20260801 (MacQueen, 1967; Arthur and Vassilvitskii, 2007). The number of clusters ranges from 2 to 12. CLASSIX uses Euclidean distance and sorting along the first principal direction. Radius ranges from 0.25 to 1.75 in increments of 0.05. Minimum cluster size takes values 1, 5 and 10, while group merging uses either distance or density. The merge scale is 1.5, tiny groups can be merged, and post allocation is disabled. Each representation has 11 K Means candidates and 186 CLASSIX candidates. A recorded boundary diagnostic triggered a symmetric radius extension from 1.80 to 2.25 for both representations. The extension tests grid sensitivity and does not replace the primary selection.

Each method and representation reports an unconstrained silhouette optimum and a deployment-acceptable result. A candidate is internally valid only when at least two nonnoise clusters remain and silhouette, Davies Bouldin and Calinski Harabasz indices are finite (Rousseeuw, 1987; Davies and Bouldin, 1979; Caliński and Harabasz, 1974). Deployment acceptability further requires 2 to 12 nonnoise clusters, no more than 20 per cent noise and no single nonnoise cluster larger than 95 per cent of nonnoise observations. Selection maximises silhouette, then prefers lower Davies Bouldin, higher Calinski Harabasz, a smaller largest cluster share, less noise, fewer clusters and finally the lexical parameter record. Runtime is measured but never selects a model. Keeping the unconstrained result reveals when an attractive internal score is produced by a partition that is unsuitable under the stated operational constraints.

Internal indices exclude observations labelled as noise, so CLASSIX index comparisons refer to each partition's nonnoise subset. Cross-representation partition comparisons retain every purchaser because a noise assignment is itself a difference in assignment. Fixed K comparisons use the same value from 2 to 12. Fixed CLASSIX comparisons use the same radius, minimum size and merge rule. Independently selected deployment partitions are compared by adjusted Rand index, arithmetic-mean normalised mutual information and variation of information (Hubert and Arabie, 1985; Meilă, 2007). A Hungarian match aligns cluster labels for transition tables (Kuhn, 1955). The resulting migration share describes changed assignments and does not imply that numeric labels identify persistent customer types.

3.2.2 Resampling and explanation evaluation

Fixed-parameter overlap-resampling stability uses ten pairs of independent 80 per cent samples without replacement. The same seed pairs are used for both representations and methods. Each sample refits its transformations and clustering with the full-data deployment parameters. Adjusted Rand index is calculated only on purchaser identifiers present in both members of a pair. The statistic measures sensitivity to sampled customers and, for K Means, initialisation, conditional on the selected configuration. It does not measure stability of model selection, calendar time or an external cohort.

The CLASSIX explanation retains public aggregation group assignments, starting point indices, final labels and cluster sizes. Merge membership is reconstructed from public group and final assignments because version 1.5.1 does not expose a documented complete edge list. Every exported provenance label is checked against the selected main result for the same visitor. The trace is an explanation of the fitted CLASSIX partition and does not claim that a group boundary is a human validated customer rule.

ExKMC approximates K Means labels under five deterministic 80 per cent training and 20 per cent test splits. The primary full cohort selection gives four K Means clusters in both representations, so K is fixed at four before splitting. Log transformations, standardisation, the K Means reference and the ExKMC tree are fitted using training rows only. The test target is the label returned by the fitted K Means predictor. This estimates explanation fidelity conditional on the primary K choice rather than nested selection of K. The leaf budget ranges from K to four times K. The main comparison fixes the budget at K, while the wider range measures the fidelity and complexity tradeoff. A supplementary analysis selects K from 2 to 12 using each training split alone. Fidelity is the proportion of exact label agreements. Complexity is reported through effective leaves, depth, path conditions and features. Rule thresholds are exported in both standardised and original units.

Business addressability is evaluated for selected CLASSIX and K Means clusters through a documented audit rather than subjective cluster names. A nonnoise cluster must contain at least 50 customers and one per cent of nonnoise customers, retain at least 80 per cent category coverage, and differ from the global median by at least half an interquartile range on two to four named features. Passing this audit shows that a cluster can be expressed as a limited set of query conditions with adequate coverage. It does not demonstrate intervention effectiveness, revenue change or human comprehension.

3.2.3 Projection filtering and complexity

Let processed observations be $x_i \in \mathbb{R}^d$, let u be a unit sorting direction and let $s_i = u^\mathsf{T} x_i$. CLASSIX orders the scores and begins a group at a starting point c. Its aggregation radius in processed coordinates is r.

Proposition 3.1. *If a later point satisfies $s_i - s(c) > r$, then it cannot satisfy $\|x_i - c\|_2 \leq r$.*

Proof. Cauchy Schwarz gives

$$|s_i - s(c)| = |u^\mathsf{T}(x_i - c)| \leq \|u\|_2 \|x_i - c\|_2 = \|x_i - c\|_2.$$

The strict score difference implies a distance greater than r. The implementation includes equality in both the score window and exact squared distance comparison, so the filter introduces no mathematical false negative apart from floating point effects. □

The filter gives exact pruning for a fixed order but does not make greedy group assignment independent of that order. If n observations produce q aggregation groups, aggregation retains at most $n(n-1)/2$ exact point comparisons. Distance merging retains at most $q(q-1)/2$ starting point comparisons. The worst case distance arithmetic is

$$O\big((n^2 + q^2)d\big)$$

in addition to principal direction estimation and scalar sorting. The package counter records retained aggregation comparisons on the observed data. It is an empirical quantity and not a proof of linear runtime.

Conditional determinism requires more than a fixed random seed. A positive leading eigengap fixes the one dimensional principal subspace, while the implementation's sign convention fixes its orientation when that convention is separated from zero. The sorted order is fixed when relevant scores are distinct. Exact ties require an explicit stable secondary order or an order invariance check. Radius, merging, minimum size and nearest eligible group decisions must also avoid unresolved equalities. These conditions distinguish deterministic execution on the observed input from a global statement about all nearby inputs.

3.2.4 A local stability certificate

Let the processed covariance matrix be S with leading eigenvalues λ_1 and λ_2. Define the eigengap $\gamma = \lambda_1 - \lambda_2 > 0$. Let a perturbation change the covariance by ΔS and align the perturbed direction $\widehat{u}$ with u. When $2\|\Delta S\|_2 < \gamma$, the Davis Kahan theorem gives the conservative bound (Davis and Kahan, 1970)

$$\sin \angle(\widehat{u}, u) \leq \frac{2\|\Delta S\|_2}{\gamma}.$$

After sign alignment, $\|\widehat{u} - u\|_2 \leq \sqrt{2} \sin \angle(\widehat{u}, u)$. If every processed point changes by at most ε_x and has norm at most B, a conservative score change bound is

$$b_s = \varepsilon_x + \sqrt{2} B \frac{2\|\Delta S\|_2}{\gamma}.$$

Let g_s be the minimum adjacent score gap and g_o the orientation rule's absolute score. Let g_w be the minimum margin of every relevant projected score difference from the window boundary r. Let g_r be the minimum margin of tested aggregation distances from r. For the distance merging branch, let g_m be the corresponding margin between every tested starting-point distance and the active merge threshold. Let b_r and b_m bound the changes of those distance tests.

Proposition 3.2. *The fitted partition under distance merging is unchanged by the perturbation if $g_s > 2b_s$, $g_o > b_s$, $g_w > 2b_s$, $g_r > b_r$ and $g_m > b_m$, provided that the representatives, eligible groups, minimum size decisions, tiny group rule, post allocation rule and all relevant tie decisions remain fixed.*

Proof. The first inequality preserves every relevant pairwise score order because each endpoint can move by at most b_s. The second preserves the package orientation sign. The third fixes window membership because each score difference moves by at most $2b_s$. The fourth preserves every accepted or rejected radius comparison, while the fifth preserves every merge comparison. With the branch, representatives and eligible set fixed, these unchanged finite decisions reproduce the same groups, merge components and final labels. □

The certificate is local and sufficient rather than necessary. It does not cover density merging, whose intersection and density decisions require separate margins. Davis Kahan controls only the principal direction. It does not preserve greedy order, aggregation, merging or noise decisions unless the additional finite margins hold. Empirical Gaussian perturbation and customer resampling are reported separately from this statement.

3.2.5 Benchmark and robustness designs

Nine labelled datasets provide an algorithm-level benchmark for CLASSIX, K Means, DBSCAN and Ward (Ester et al., 1996; Ward, 1963). Candidate selection uses only internal indices and deployment guards. Reference labels are reserved for ARI and NMI evaluation. K Means results average five fixed seeds before parameter selection. Ward, DBSCAN and CLASSIX store one structural row per candidate, while repeated fits audit label identity and runtime on the recorded environment. This empirical identity check is not a global determinism theorem.

The rich representation undergoes twelve leave-one-feature-out fits. PCA retains at least 90 per cent of standardised variance and is compared with the named space on structure and stability, never on rule quality (Pearson, 1901; Hotelling, 1933). CLASSIX perturbation uses standard deviations 0, 0.000001, 0.0001, 0.001 and 0.01 with ten seeds at every nonzero level. Session counts are rebuilt with gaps of 15, 30 and 60 minutes. A negative control appends independently permuted rich-only dimensions to unchanged compact features. These diagnostics retain the primary selected configurations unless a section explicitly reports complete feature reselection.

Artificial intelligence tools assisted with code drafting, data processing, numerical checks and source discovery. The author reran the analyses from saved code, verified retained sources and checked reported values against recorded outputs. Reproducibility details are supplied with the accompanying materials.

Chapter 4

Results and Analysis

The results follow the evidence chain established in Chapter 3. A benchmark first checks the implementations and selection procedure. The customer analysis then examines fitted partitions, explanation and addressability before sensitivity tests probe the main observations.

4.1 Algorithm benchmark

Nine labelled datasets provide an algorithm check before the customer analysis. Six shape datasets and three multivariate datasets were evaluated with CLASSIX, K Means, DBSCAN and Ward. Selection used internal indices and deployment guards. Reference labels entered only the subsequent ARI and NMI calculation.

No method dominates the benchmark. K Means has the highest mean ARI across the nine selected results at 0.597, followed by Ward at 0.575, DBSCAN at 0.509 and CLASSIX at 0.508. CLASSIX and Ward share a mean rank of 2.44, while K Means has 2.39 and DBSCAN has 2.72. These averages conceal marked dependence on shape. CLASSIX recovers the Jain reference exactly but reaches only 0.264 ARI on R15. DBSCAN recovers Spiral exactly and performs best on Aggregation and Compound, yet its selected Wine candidate has no valid internal score. K Means is strongest on Seeds and Wine, while Ward is strongest on R15.

Observed runtime depends on implementation and dataset size. Median selected runtime is 0.011 seconds for CLASSIX, 0.064 seconds for K Means, 0.003 seconds for DBSCAN and 0.004 seconds for Ward on the recorded machine. These values do not establish a universal speed ordering. The benchmark instead verifies that the implementations, label-free selector and external evaluation recover varied successes and failures before the same metric code is applied to RetailRocket. The customer conclusion rests on paired evidence rather than algorithm superiority.

Table 4.1 Highest external ARI after label free selection on each benchmark dataset

Dataset	Method	ARI
Aggregation	DBSCAN	0.910
Compound	DBSCAN	0.826
Iris	CLASSIX and K Means	0.568
Jain	CLASSIX	1.000
Pathbased	CLASSIX	0.497
R15	Ward	0.804
Seeds	K Means	0.481
Spiral	DBSCAN	1.000
Wine	K Means	0.897

4.2 Representation and customer cluster structure

Changing the represented information alters the selected customer structure for both clustering methods. Table 4.2 reports the independently selected deployment-acceptable configurations. K Means selects four clusters in each representation, but the agreement between the two four-cluster partitions is only 0.014 ARI. CLASSIX selects two compact clusters and three rich nonnoise clusters with 218 additional noise assignments. Their cross-representation agreement is 0.068 ARI. These values are close to the chance-adjusted baseline even though the customer identifiers and algorithm families are unchanged.

Table 4.2 Deployment acceptable clustering results on the shared purchaser cohort

Representation	Method	Clusters	Noise	Largest share	Silhouette	Stability
Compact	CLASSIX	2	0	0.909	0.595	0.985
Rich	CLASSIX	3	218	0.898	0.291	0.965
Compact	K Means	4	0	0.664	0.560	0.988
Rich	K Means	4	0	0.579	0.351	0.995

The internal indices do not move in a single direction. Compact CLASSIX has silhouette 0.595, Davies Bouldin 1.057 and Calinski Harabasz 4285.5. The rich result has silhouette 0.291, Davies Bouldin 0.859 and Calinski Harabasz 1516.9. These CLASSIX values are calculated on each partition's nonnoise observations. Davies Bouldin favours the rich result, while silhouette and Calinski Harabasz favour the compact result. Compact K Means has silhouette 0.560 and Calinski Harabasz 10459.1, compared with 0.351 and 3923.4 in the rich space. Davies Bouldin rises from 0.775 to 1.040. Richer behaviour produces a different geometry whose apparent quality depends on the criterion.

The deployment rule materially changes the result that would be chosen by silhouette alone. Compact CLASSIX reaches silhouette 0.899 by placing 11,698 of 11,719

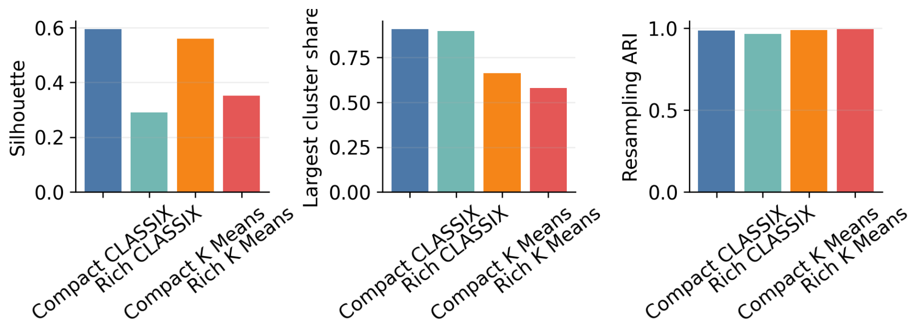

Figure 4.1 Internal structure and resampling stability for the selected customer partitions

customers in one cluster. Rich CLASSIX reaches 0.792 with 11,716 customers in one cluster and the remaining three customers split across two tiny clusters. Compact K Means reaches 0.850 with 11,593 customers in one of two clusters. All three partitions fail the maximum share rule. The deployment selections sacrifice these high silhouette values to avoid reporting a nearly undivided customer base as a useful segmentation. Rich K Means is the only case where the unconstrained and deployment selections coincide.

Low agreement is not caused only by choosing different numbers of clusters. At every fixed K from 2 to 12, the K Means ARI between compact and rich representations remains between 0.012 and 0.113. At the common selected value of four it is 0.014, with NMI 0.089 and variation of information 1.691 nats. Under matched labels, 54.61 per cent of customers move to a different K Means cluster. Among 41 CLASSIX settings that are deployment acceptable in both spaces, matched-parameter ARI ranges from -0.051 to 0.392 with a median of -0.023. The independently selected CLASSIX partitions have NMI 0.054 and variation of information 0.693 nats, with 18.66 per cent assigned to a different matched cluster or noise state. The lower CLASSIX migration share partly reflects the dominance of one large cluster in both representations and should not be read as stronger substantive agreement.

Cluster sizes make the changes more concrete. Compact K Means contains 7,784, 2,494, 1,344 and 97 customers. Rich K Means contains 6,790, 3,607, 1,156 and 166. Its 166 customer cluster has a median of 338.5 total events, 28 sessions and 12 top categories, whereas the full cohort medians are 6 events, 2 sessions and 1 top category. Another rich cluster of 1,156 customers has no known category at its median and a purchase recency median of 127.4 days. These are descriptive contrasts in observed features, not customer identities or causal segments.

CLASSIX yields a sharper imbalance. The compact deployment result contains 10,656 and 1,063 customers. The rich result contains 10,331, 1,137 and 33 nonnoise customers plus 218 noise assignments. The 33-customer cluster is highly active, with medians of 869 total events, 36 sessions and 14 top categories. Its small size fails the documented addressability size rule. The 1,137-customer cluster has zero category coverage and fails the coverage rule. The dominant 10,331-customer cluster has no two sufficiently distinctive median effects. Richer features expose unusual behavioural combinations, but the selected CLASSIX partition does not turn them into a balanced set of directly addressable clusters.

Fixed-parameter overlap-resampling indicates that these structural differences are reproducible under the stated sampling scheme. Mean overlap ARI is 0.985 for compact CLASSIX and 0.965 for rich CLASSIX. The corresponding K Means values are 0.988 and 0.995. The lowest of ten pairwise values is 0.881 for compact CLASSIX and 0.941 for rich CLASSIX. Stable refitting does not imply that a partition is uniquely correct or that selection would recur. It shows that each selected representation and parameter combination tends to reproduce its own assignment pattern when one fifth of the cohort is omitted.

4.3 Explanation complexity, fidelity and addressability

The explanation results separate provenance from approximation. CLASSIX exposes how its own observations are aggregated and merged. ExKMC approximates a K Means predictor with axis aligned rules. Their numbers answer different questions and are not combined into a common explanation score.

The selected compact CLASSIX fit produces 131 aggregation groups that merge into two final clusters. The rich fit produces 559 groups and four final states consisting of three clusters and noise. Every exported customer trace contains the visitor identifier, aggregation group, representative row and final cluster. Both exports reproduce the selected full data labels exactly with ARI 1. Rich CLASSIX needs more than four times as many local aggregation groups even though its final nonnoise cluster count rises only from two to three. Group count describes computational provenance rather than reading difficulty.

The main ExKMC comparison fixes K at four for both representations because both primary K Means selections contain four clusters. At the four-leaf budget, both trees use four effective rules with depth 3 and 2.25 conditions per rule. Compact mean training fidelity is 0.9466 and mean held-out fidelity is 0.9448. Rich fidelity is higher at 0.9639 and 0.9650. Across the five splits, compact held-out fidelity ranges from 0.9416 to 0.9480 and rich fidelity ranges from 0.9599 to 0.9697. This is descriptive split variability rather than inferential uncertainty. The close training and test values give no sign of a training-only gain that disappears on held-out customers.

Compact rules use purchase occasion count and purchase recency across all five splits. Rich rules use purchase occasion count, activity span and the number of distinct top categories. Thresholds are exported in original units after reversing the training scaler and logarithmic transform. Named thresholds support direct database style conditions, but they inherit the meaning and missingness limits of the constructed variables.

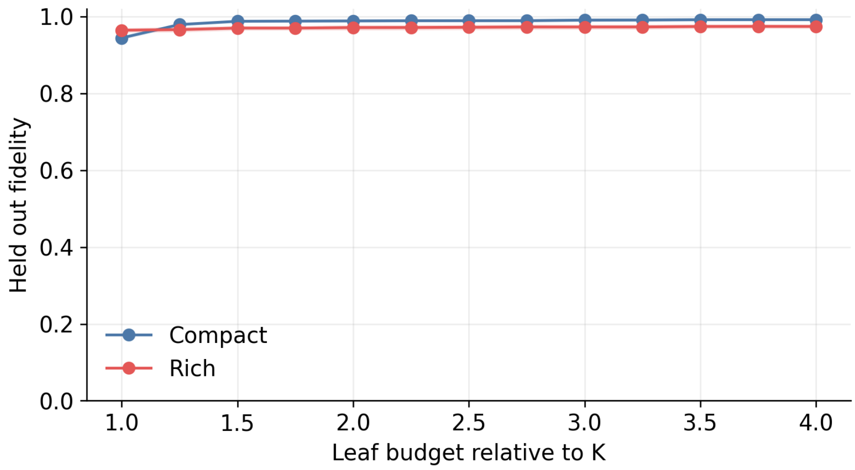

Figure 4.2 Held out ExKMC fidelity across the fixed leaf budget

Additional leaves change the two fidelity frontiers differently. At sixteen leaves, compact mean held out fidelity reaches 0.9924 with mean depth 6.6 and 4.84 conditions per rule. Rich fidelity begins higher at four leaves but reaches 0.9747 with mean depth 6.4 and 4.74 conditions at sixteen leaves. Compact fidelity begins lower and gains more from expansion. A supplementary training only selection chooses two compact clusters and four rich clusters in every split. Its compact fidelity is 0.9985 with two rules, but that result is not used for the main complexity comparison because K differs.

The addressability audit is stricter than rule export. Neither compact CLASSIX cluster passes because neither differs from the cohort median on at least two named features by the required half interquartile range. In the rich result, the 33-customer cluster is too small, the 1,137-customer cluster has zero category coverage, and the 10,331-customer cluster lacks two sufficiently distinctive median conditions. None of the five nonnoise CLASSIX clusters is classified as addressable. No compact K Means cluster passes. One of four rich K Means clusters passes all gates, while the others fail because they have too many distinguishing conditions or insufficient category coverage.

Business interpretation remains narrow. Rich features expose a high-activity minority and a group with missing category histories. ExKMC shows that K Means assignments can be represented with a small number of named thresholds at high held-out fidelity. The one passing rich cluster meets a technical audit rather than an outcome test. The observed value is limited to queryability and auditable coverage. Revenue effect, campaign response and human comprehension remain unmeasured.

4.4 Sensitivity and robustness

The representation effect is not tied to one diagnostic. Its interpretation changes when the analysis preserves selected parameters, repeats selection after changing features, or destroys joint information. Fixed checks measure local dependence. Reselection repeats the candidate search and deployment rule. Permutation tests added dimensions with no aligned customer information.

4.4.1 Named features and reduced geometry

Fixed parameter deletion identifies category behaviour as the strongest local influence on the rich partitions. Removing distinct top categories reduces agreement with the full rich result to ARI 0.189 for CLASSIX and 0.516 for K Means. Removing top category share gives ARI 0.238 and 0.517. Every other CLASSIX deletion remains between 0.927 and 0.967. The corresponding K Means values remain between 0.855 and 0.991. Activity span and mean interevent time have the next largest effect on K Means, with ARI 0.874 and 0.855. Influence is unevenly distributed across the named variables, but category information alone does not generate a preferable segmentation.

The PCA diagnostic preserves 91.32 per cent of standardised rich variance in six components. It is a fixed-parameter local diagnostic rather than PCA-space model selection. K Means in this reduced space agrees with the named rich partition at ARI 0.995 and has silhouette 0.390 compared with 0.351. CLASSIX retains three clusters, lowers noise from 218 to 71 customers and agrees with the named result at ARI 0.920. Its silhouette is 0.304 compared with 0.291. Mean resampling ARI is 0.995 for reduced K Means and 0.987 for reduced CLASSIX. The principal component space preserves most fitted structure. It is not used for explanation because rotated components do not retain the direct business meaning of the twelve named variables.

Complete reselection confirms that category information creates the largest structural turn. All twenty two scenarios produce a deployment acceptable candidate. Removing distinct top categories changes CLASSIX to eleven clusters with ARI 0.153 against the full rich partition. K Means changes to two clusters with ARI 0.728. Removing the whole category block produces seven CLASSIX clusters with ARI 0.126 and two K Means clusters with ARI 0.729. The progressive endpoints reproduce the compact and rich primary partitions exactly. These 4,334 candidate fits show that the category effect persists when parameters and cluster counts are allowed to change.

CLASSIX sensitivity is distributed across the grid rather than concentrated at one radius. Deployment acceptable candidates occur at twenty three of thirty one compact radii from 0.35 to 1.45 and twenty rich radii from 0.30 to 1.25. Minimum size 1 produces no

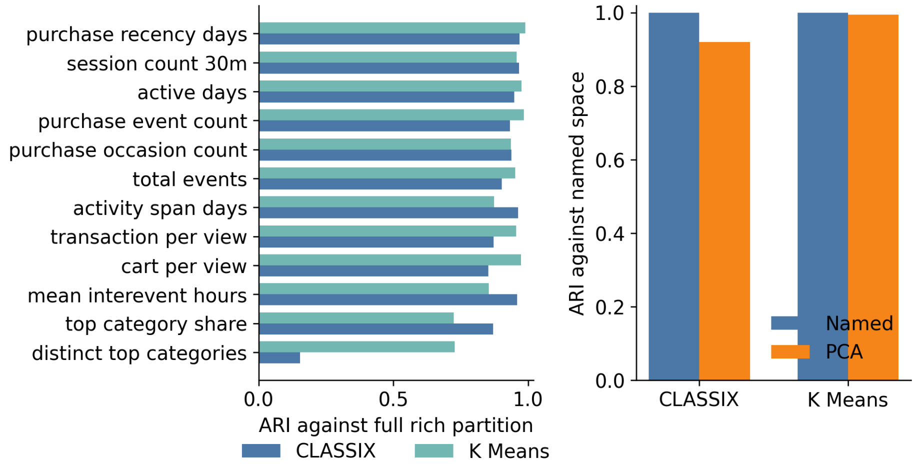

Figure 4.3 Reselected feature deletion and PCA diagnostics

acceptable candidate in either representation. Minimum sizes 5 and 10 produce 23 and 32 acceptable compact candidates, compared with 19 and 27 rich candidates.

4.4.2 Perturbation and temporal assumptions

Small Gaussian perturbations leave the selected CLASSIX partitions almost unchanged. Compact agreement remains ARI 1 through a noise standard deviation of 0.001. At 0.01 the mean ARI is 0.999 and the minimum is 0.996. Rich agreement is also exact through 0.0001. It falls to a mean of 0.995 at 0.001 and 0.985 at 0.01, with minima of 0.985 and 0.962. The richer fit is consequently more sensitive to local numerical changes under the tested scale, but it does not become unstable in the sense of wholesale reassignment. These empirical results do not certify the density merging branch covered by the rich selection. The mathematical certificate applies only to distance merging when its stated margins hold.

Reconstructing sessions from raw events gives a similar qualified result. The selected thirty minute window reproduces both partitions exactly. A fifteen minute window gives ARI 0.976 for K Means and 0.971 for CLASSIX. A sixty minute window gives ARI 0.985 and 0.959. The four cluster K Means result and the three cluster CLASSIX result persist at both alternative windows, although the CLASSIX noise count changes from 218 to 220 and 239. The conclusions do not arise solely from choosing exactly thirty minutes, while the detailed assignments still depend on the session convention.

4.4.3 Dimensionality control

The permutation control sets a stricter boundary on causal interpretation. The compact columns remain unchanged while the nine additional dimensions are independently permuted across customers. These columns preserve their marginal distributions but destroy their joint relationship with the compact representation and with each other. With the compact K Means setting held fixed, adding all nine permuted columns lowers agreement with the compact partition to ARI 0.017 and silhouette to 0.127. With the compact CLASSIX setting held fixed, adding even one permuted dimension produces a single cluster and ARI 0.

The single deterministic permutation control is an existence demonstration rather than a variance decomposition. Standardisation gives each added variable comparable weight, so random added dimensions can alter distances, principal directions and radius neighbourhoods in at least one realised order. The control does not estimate how much of the observed compact-to-rich difference is caused by dimensionality. Named feature deletion and PCA separately show that category and temporal information contribute reproducible structure in the observed rich representation.

4.5 Integrated discussion and validity limits

The study concerns purchasers recorded in one anonymous event log. It excludes visitors without a transaction event and has no demographic, price, revenue, campaign or outcome data. Category features cover 76.16 per cent of events, while 1,168 purchasers have no known category. Zero ratios can indicate a genuine absence of views or the limits of recorded behaviour. These boundaries prevent substantive labels such as valuable, loyal or profitable from being inferred from the clusters.

Internal indices and fixed-parameter overlap-resampling measure geometric separation and conditional repeatability within the observed design. They do not establish an objectively correct number of customer groups or stability of complete model selection. The deployment constraints prevent nearly single-cluster solutions from dominating selection, but the thresholds are operational choices rather than universal standards. The nine labelled benchmarks provide algorithm checks across several shapes and dimensions, not a guarantee that their relative performance transfers to customer data.

Explanation evidence has separate limits. CLASSIX provenance is exact for its own fitted labels but does not produce a short business rule. ExKMC fidelity measures agreement with a K Means predictor and does not validate that predictor. The addressability audit checks size, coverage and observable distinctiveness, yet none of the selected CLASSIX clusters passes it. Business value here means the capacity to inspect provenance, export named conditions and audit coverage. Intervention response, commercial return and human understanding require prospective evaluation beyond the available data.

Chapter 5

Conclusion

This dissertation examined how changing the information used to represent the same 11,719 purchasers affects observable and explainable cluster structure. Compact and rich representations produce substantially different assignments for both K Means and CLASSIX. The compact space gives higher silhouette values, while the rich space gives a more balanced K Means partition and many more CLASSIX aggregation groups. Independently selected partitions disagree, and matched CLASSIX settings also give low or variable agreement. Fixed-parameter overlap-resampling remains high within each representation, so conditional repeatability does not imply agreement across representations or stability of model selection.

The explanation results reveal a representation-dependent frontier. At four leaves, compact ExKMC fidelity is 0.9448 and rich fidelity is 0.9650 with equal tree complexity. At sixteen leaves, compact fidelity rises to 0.9924 while rich fidelity reaches 0.9747. CLASSIX provenance remains exact, yet the rich fit requires 559 aggregation groups compared with 131. One rich K Means cluster passes the addressability audit, while no compact K Means or CLASSIX cluster passes. Richer behaviour increases descriptive reach without guaranteeing simpler explanations, superior internal scores or deployment-ready groups.

Representation is not a neutral preprocessing choice. It determines the geometry against which clustering quality, stability and explainability are judged. The benchmark establishes that the implementations and metric path behave across varied labelled structures, while the local certificate states only the margins sufficient for the distance-merging branch. Future work should test whether representation-dependent groups predict prospective responses under complete category and value data. It should also examine whether analysts understand and act on the exported provenance and rules. Until then, technical queryability is not realised business value.

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Appendix A

Technical Reproduction Map

The reported experimental results are reproduced from the distributed data with `python-msrc.reproduce`. The fixed sequence covers the benchmark, customer clustering, explanations, sensitivity analyses, summary tables and figures. The optional `--compile-pdf` flag also rebuilds the report. Environment versions and random seeds are recorded in `environment.yml` and `protocol/analysis_protocol.json`.

Primary clustering selections are stored in `results/retailrocket/selected_configurations.csv`. Customer level labels, transitions, profiles and resampling records remain separate so that every reported aggregate can be recalculated. ExKMC training selection, held out fidelity and exported rules are stored in distinct files. CLASSIX provenance includes an exact label audit against the selected result.

The sensitivity directory separates fixed parameter diagnostics from complete candidate reselection. PCA records that reduced components are not permitted for rules. The zero perturbation rows have ARI 1. Session windows are reconstructed from the distributed purchaser timestamp table. Permuted dimensions retain the compact columns exactly while destroying their joint customer alignment.

The distributed feature tables and purchaser event timestamps are sufficient for the reported experiments. The optional `--rebuild-features` flag reconstructs these tables when the four original RetailRocket files are available. The repository `README.md` records the execution sequence, output map and licence boundary.